

Linux RAG Dataset - PDF 3: Permissions & User Management

chmod

Category: Permissions

Description: Change file permissions

Syntax: chmod [OPTIONS] MODE FILE

Example: chmod 755 script.sh

Options: -R recursive, -v verbose

Common Error: Permission denied

Solution: Check ownership or use sudo

Common Mistake: Using 777 unnecessarily

Related: chown

Best Practice: Secure scripts

Difficulty: Intermediate

chown

Category: Permissions

Description: Change file owner

Syntax: chown USER:GROUP FILE

Example: chown alice:dev file.txt

Options: -R recursive

Common Error: Operation not permitted

Solution: Use sudo

Common Mistake: Changing system files

Related: chmod

Best Practice: Set correct ownership

Difficulty: Intermediate

chgrp

Category: Permissions

Description: Change group ownership

Syntax: chgrp GROUP FILE

Example: chgrp dev project

Options: -R recursive

Common Error: Invalid group

Solution: Create group first

Common Mistake: Wrong group

Related: groups

Best Practice: Use project groups

Difficulty: Beginner

umask

Category: Permissions

Description: Set default permissions

Syntax: umask MODE

Example: umask 022

Options:

Common Error: Unexpected permissions

Solution: Verify mask

Common Mistake: Too restrictive mask

Related: chmod

Best Practice: Set organization policy

Difficulty: Intermediate

sudo

Category: Users

Description: Run as superuser
Syntax: sudo COMMAND
Example: sudo apt update
Options: -u user
Common Error: User not in sudoers
Solution: Edit sudoers safely
Common Mistake: Running everything as root
Related: visudo
Best Practice: Use least privilege
Difficulty: Beginner

su

Category: Users
Description: Switch user
Syntax: su USER
Example: su root
Options: -l login shell
Common Error: Authentication failure
Solution: Correct password
Common Mistake: Staying as root
Related: sudo
Best Practice: Exit after work
Difficulty: Beginner

passwd

Category: Users
Description: Change password
Syntax: passwd USER
Example: passwd alice
Options:
Common Error: Authentication token error
Solution: Retry as root
Common Mistake: Weak passwords
Related: chage
Best Practice: Use strong passwords
Difficulty: Beginner

useradd

Category: Users
Description: Create new user
Syntax: useradd USER
Example: useradd bob
Options: -m create home
Common Error: User exists
Solution: Choose another name
Common Mistake: No home directory
Related: usermod
Best Practice: Create home
Difficulty: Intermediate

userdel

Category: Users
Description: Delete user
Syntax: userdel USER
Example: userdel bob
Options: -r remove home
Common Error: User logged in
Solution: Log out first

Common Mistake: Deleting wrong account

Related: useradd

Best Practice: Backup data

Difficulty: Intermediate

usermod

Category: Users

Description: Modify user

Syntax: usermod OPTIONS USER

Example: usermod -aG sudo bob

Options: -aG append group

Common Error: Group missing

Solution: Create group

Common Mistake: Overwriting groups

Related: groupadd

Best Practice: Use -a with -G

Difficulty: Intermediate

groupadd

Category: Groups

Description: Create group

Syntax: groupadd GROUP

Example: groupadd developers

Options:

Common Error: Group exists

Solution: Rename group

Common Mistake: Duplicate names

Related: groupmod

Best Practice: Consistent names

Difficulty: Beginner

groupdel

Category: Groups

Description: Delete group

Syntax: groupdel GROUP

Example: groupdel temp

Options:

Common Error: Primary group

Solution: Reassign users

Common Mistake: Deleting active group

Related: groupmod

Best Practice: Check members

Difficulty: Intermediate

groupmod

Category: Groups

Description: Modify group

Syntax: groupmod OPTIONS GROUP

Example: groupmod -n dev developers

Options: -n rename

Common Error:

Solution:

Common Mistake: Renaming active groups

Related: groupadd

Best Practice: Plan changes

Difficulty: Intermediate

groups

Category: Groups

Description: Show user groups

Syntax: groups USER

Example: groups alice

Options:

Common Error:

Solution:

Common Mistake:

Related: id

Best Practice: Verify memberships

Difficulty: Beginner

id

Category: Users

Description: Display UID/GID

Syntax: id USER

Example: id alice

Options:

Common Error:

Solution:

Common Mistake:

Related: groups

Best Practice: Debug permissions

Difficulty: Beginner

whoami

Category: Users

Description: Current username

Syntax: whoami

Example: whoami

Options:

Common Error:

Solution:

Common Mistake:

Related: id

Best Practice: Useful in scripts

Difficulty: Beginner

who

Category: Users

Description: Logged in users

Syntax: who

Example: who

Options:

Common Error:

Solution:

Common Mistake:

Related: w

Best Practice: Monitor sessions

Difficulty: Beginner

w

Category: Users

Description: Show logged users and activity

Syntax: w

Example: w

Options:

Common Error:

Solution:

Common Mistake:

Related: who

Best Practice: Check load

Difficulty: Beginner

last

Category: Users

Description: Login history

Syntax: last

Example: last

Options:

Common Error:

Solution:

Common Mistake:

Related: who

Best Practice: Audit logins

Difficulty: Intermediate

visudo

Category: Admin

Description: Edit sudoers safely

Syntax: visudo

Example: visudo

Options:

Common Error: Syntax error

Solution: Use visudo only

Common Mistake: Editing file directly

Related: sudo

Best Practice: Always validate

Difficulty: Advanced